

	D	C	B	A
Task A	1. Implements basic clustering on at least 3 of the 5 required features (Distance, Fare, Tip, Rate-code, Pickup zone) 2. Uses only one algorithm (k-means or hierarchical) 3. Minimal distance metric justification 4. No cluster validation	1. Implements clustering on all 5 required features 2. Tests at least 2-3 different distance metrics (e.g., Euclidean, Manhattan) 3. Some cluster validation 4. Some interpretation of clusters	1. All of C; and 2. Thoughtful data processing, including feature scaling and handling of categorical variables (rate-code, pickup zone) 3. Clear justification for distance metric selection based on feature characteristics (e.g. Euclidean for continuous and appropriate ones for others)	1. All of B; and 2. Discuss tradeoffs in cluster-quality metrics 3. Explore different values of 'k' systematically
Task B	1. Correctly applies best performing model from previous submission 2. Transfer of hyperparameters appropriately	1. All of D; and 2. Calculates standard performance metrics (e.g., accuracy, F-1 score, etc.)	1. All of C; and 2. Calculates confusion matrix	1. All of B; and 2. Address class imbalance, if present, in payment_type distribution
Task C	1. Maps cluster centroids to payment_type label 2. Basic comparison of clustering and supervised learning performance	1. All of D; and 2. Discusses which approach performs better with evidence	1. All of C; and 2. Good mapping of cluster centroids to payment_type label (e.g., using majority voting) 3. Presents comparison results clearly	1. All of B; and 2. Discusses why clustering might succeed or fail, compared to supervised learning 3. Well organized Jupyter notebook with clear narrative flow